Andrea Volpi

andreavolpi207801@gmail.com Krakowskie Przedmieście 3, 00-927 Warszawa, Poland Homepage
ORCID

Personal information

Date of birth: January 23, 1996

Place of birth: Brescia, Italy

Languages: Italian (native), English (fluent), German (B1)

Research interest

Reverse mathematics; Computability theory; Ramsey theory; Ordinal analysis

Education

Ph.D.:	Mathematics, University of Udine, Italy (supervisor: prof. Alberto Marcone)	Nov 2022 to Mar 2026
	Visiting scholar at UConn, Storrs, Connecticut, US (hosts: prof. Damir Dzhafarov, prof. Reed Solomon)	Jan 2023 to Apr 2023
	Visiting scholar at NUS, Singapore (host: dr. Jun Le Goh)	Sep 2025 to Oct 2025
M.Sc.:	Mathematics (LM-40), University of Trento, Italy	Jan 2019 to May 2021
B.Sc.:	Mathematics (L-35), University of Trento, Italy	Aug 2015 to Nov 2018

Employment

Postdoc:	Faculty of philosophy, University of Warsaw, Poland "Structural complexity measures for foundational theories" P.I. Mateusz Łełyk	Apr 2026 to present

Graduate schools

Summer School on Proof Theory	Ghent, Belgium Sep 2025 1 week
Reverse mathematics: new paradigms ESI	Vienna, Austria Jul 2025 to Aug 2025 1 week
IMS graduate summer school in logic	Singapore Jun 2023 to Jul 2023 3 weeks
Computability and combinatorics summer school 2023	Hartford, Connectictut May 2023 1 week
Discrete mathematics and logic: between mathematics and the computer science	Marseille, France Jan 2023 1 week

AILA logic summer school — Gargnano, Italy, Aug 2022, 1 week

Teaching experience

Teaching assistant, University of Udine, Italy:

- Geometry I, B.Sc. in mathematics, Sep 2024 to May 2025
- Linear algebra, B.Sc. in engineering, Sep 2024 to Dec 2024
- Calculus I, B.Sc. in mathematics, Nov 2023 to Dec 2023
- Calculus I, B.Sc. in informatics, Sep 2023 to Oct 2023
- Calculus I, B.Sc. in informatics and IBML, Jan 2023 to Sep 2023

Teaching assistant, University of Trento, Italy:

- Geometry and linear algebra, B.Sc. in engineering, Sep 2020 to Dec 2020
- Geometry and linear algebra, B.Sc. in engineering, Jan 2020 to May 2020
- Geometry I, B.Sc. in mathematics, Sep 2019 to Dec 2019
- Geometry and linear algebra, B.Sc. in engineering, Jan 2019 to May 2019

Publications and ongoing research

[1] **The barrier Ramsey theorem**, joint with Alberto Marcone and Antonio Montalbán, arxiv:2505.02544 (accepted)

[2] **Reverse mathematics and dimension of posets**, joint with Alberto Marcone, arxiv:2603.18759 (accepted)

[3] **Effectiveness and strong graph indivisibility**, joint with Damir Dzhafarov and Reed Solomon, arxiv:2411.16950 (submitted)

[4] **The strength of Ramsey theorem for α-large sets**, joint with Lorenzo Carlucci and Konrad Zdanowski, arxiv:2603.22579 (submitted)

[5] **Classification of ultrahomogeneous graphs and posets**, joint with Jun Le Goh, (work in progress)

Conferences attended

Workshop on Proof Theory Ghent, Belgium — Sep 4-5, 2025

Computability in Europe 2025 Lisbon, Portugal — Jul 14-18, 2025

Reverse mathematics ESI Vienna, Austria — 23 Jun-29 Aug, 2025

Weihrauch Complexity: Structuring the Realm of Non-Computability Schloss Dagstuhl, Germany — Mar 23-28, 2025

XXVIII AILA logic meeting 2024 Udine, Italy — Sep 3-6, 2024

Models, sets and classifications PRIN meeting 2024 Udine, Italy — Sep 2, 2024

Leeds computability days 2024 Leeds, UK — Jul 2-5, 2024

Logic colloquium 2024 Gothenburg, Sweden — Jun 24-28, 2024

RaTLoCC Pisa, Italy — Jun 11-14, 2024

Computability and combinatorics conference 2023 Hartford, Connecticut — May 19-21, 2023

Selected talks

Sapienza LOC3 seminar University La Sapienza, Roma (Invited) — Nov 7, 2025
Presentation title: Effectiveness and strong graph indivisibility

Singapore logic seminar National University of Singapore, Singapore (Invited) — Sep 17, 2025
Presentation title: Effectiveness and strong graph indivisibility

Proof society 2025 University of Ghent, Belgium — Sep 4, 2025
Presentation title: The barrier Ramsey theorem
Winner of the best student presentation award of Proof Society 2025

Computability in europe 2025 University of Lisbon, Portugal — Jul 18, 2025
Presentation title: The strength of Ramsey theorem for α-large sets

Logic working group University of Udine, Italy — Feb 21, 2025
Presentation title: Effectiveness and strong graph indivisibility

XXVIII AILA logic meeting 2024 University of Udine, Italy — Sep 5, 2024
Presentation title: Reverse mathematics and dimension of posets

Models, sets and classifications PRIN meeting University of Udine, Italy — Sep 2, 2024
Presentation title: Classifying the pigeonhole property

Logic colloquium 2024 University of Gothenburg, Sweden — Jun 26, 2024
Presentation title: Largeness notions

Logic working group University of Udine, Italy — Jun 14-21, 2024
Presentation title: Transfinite Ramsey theorem

Connecticut logic seminar UConn, Storrs, Connecticut, US (Invited) — Feb 20, 2024
Presentation title: Reverse mathematics and dimension of posets

Computability seminar CUNY graduate center, New York, US (Invited) — Feb 16, 2024
Presentation title: Largeness notions

Logic working group University of Udine, Italy — May 12, 2023
Presentation title: Reverse mathematics and dimension of posets

Organizing activities

World logic day "Le macchine di Turing" University of Udine, Italy — Jan 14, 2025
Local organizing committee

XXVIII AILA logic meeting University of Udine, Italy — 3-6 Sep, 2024
Local organizing committee

World logic day "Contare l'infinito" University of Udine, Italy — Jan 13, 2023
Local organizing committee

Logic working group "Sets and computability in Udine", University of Udine, Italy — Dec 2022 to Oct 2025
Website management

Memberships

Proof Society

CiE Computability in Europe

AILA Associazione italiana di logica e sue applicazioni

UMI Unione matematica italiana

ASL Association for symbolic logic

Awards and grants

Proof Society 2025, Ghent, Belgium: Best student presentation award (EUR 1000)

ASL Student travel grant 2025, Ghent, Belgium: Proof Society 2025 (USD 250)

ASL Student travel grant 2024, Gothenburg, Sweden: Logic Colloquium 2024 (EUR 300)

Bando di mobilità 2023-2024, Udine, Italy: Research visit to UConn (EUR 1500)